Review: Slope as rate of change

1

a No

b Yes

c No

d No

2 Run

3 The ratio of the vertical change to the horizontal change for any two points on the line.

4

a

i 2 ii 4 iii $\frac{1}{2}$

b

i -4 ii 1 iii -4

5

a Positive

b Negative

c Zero

d Undefined

6 Positive

7

a No

b Yes

c No

d No

8 13.2 cm

9 890 m

10

a $y = 1.27x$

b $\$1.27$

c $\$92.71$

d The unit rate of cost of gas per gallon.

11

a 1.4 m/min

b $y = 1.4x$

c The change in depth per minute.

d 8.4 m

e 9 minutes

12 -17 billion liters per month

13

a $\frac{6}{5}$ fish/hour

b 6 fish are caught every 5 hours.

c Slope

14

a

Days	3	6	9	12
Dollars saved	10	20	30	40

b By comparing the unit rate. The higher unit rate would be the steeper graph, which represents the person who is saving more quickly.

c David

Additional questions

15

a
i Increasing
ii 1

b
i Decreasing
ii -2

16

a
i Domain: $-2 < x \leq 4$
ii Range: $-5 < y \leq 1$
iii x-intercept is 3
y-intercept is -3
iv Rate of change is 1

b
i Domain: all real values
ii Range: all real values
iii x-intercept is 3
y-intercept is 6
iv Rate of change is -2

c
i Domain: all real values
ii Range: $y = 2$
iii There is no x-intercept
y-intercept is 2
iv Rate of change is 0

d
i Domain: $-2 \leq x \leq 4$
ii Range: $-2 \leq y \leq 1$
iii x-intercept is 2
y-intercept is -1
iv Rate of change is $\frac{1}{2}$

17

- **a** The number of bricks Neville and Charlie can lay in one minute.
- **b** Charlie
- **c** Since the time taken for laying bricks only begins when the construction workers lay their first brick, both lines must pass through the point $(0, 0)$ representing that they have laid 0 bricks after 0 minutes.

18 Alicia is mixing up the y-intercept and the rate of change. Since $f(x)$ has a greater y-intercept than $g(x)$, it is higher on the given graph. However, the rate of change of $f(x)$ is 1, while the rate of change of $g(x)$ is 2. She should have concluded that the rate of change is greater in $g(x)$ than $f(x)$ because it is steeper on the given graph.

19

- **a** Slope $= 2/45$
- **b** The slope of the line represents Natalia's speed during the open water swim, in miles per minute.

20

- **a** Slope $= 9/4$
- **b** The slope of the line represents how many batches of cookies the factory can make per hour.